

AI Exploratory Test Report

Target Application: <https://xploresyte.lovable.app>

Report Date: March 23, 2026 at 22:36

Tool Version: AI Exploratory Test Engine v2.0

Pages Analyzed: 9

Total Test Cases Generated: 53

Valid Test Cases: 52

Rejected (Hallucinated): 1

Tests Executed: 51

Passed: 25

Failed (Bugs Found): 26

2. Execution Results Summary

25 PASSED

Tests behaved correctly

26 FAILED

Bugs found in application

Bugs Found - Failed Test Cases:

Test ID	Goal	Expected (Not Met - Bug)
VT-01	Break login with empty password	validation failure
VT-03	Contaminate contact form with special characters	validation failure
VT-05	Break login with whitespace-only email	validation failure
VT-09	Exceed min limit in contact form	error message
VT-10	Test form submission with empty email field	Validation failure for missing email
VT-11	Test form submission with empty password field	Validation failure for missing password
VT-12	Test form submission with valid data and then resubmit	Validation failure or error message
VT-16	Test form submission with empty phone number field	Validation failure for missing phone number
VT-17	Test form submission with empty name field	Validation failure for missing name
VT-18	Test form submission with empty email field	Validation failure for missing email
VT-19	Break the login page with minimum password length	Password is too short error message
VT-21	Test the contact page with zero phone number	Invalid phone number error message

VT-23	Test the login page with maximum allowed password lengt	Password is too long error message
VT-25	Break the login page with one above maximum password le	Password is too long error message
VT-27	Test rapid form submission on login page	Validation failure due to duplicate submissio
VT-28	Test form submission with empty optional fields on cont	Error message for missing phone number
VT-30	Test form submission with cleared fields on login page	Validation failure due to incorrect credentia
VT-33	Test multiple submit clicks on contact page	Error message for duplicate submission
VT-34	Test form submission with rapid input filling on login	Validation failure due to incorrect credentia
VT-38	Test mode switching with incomplete fields on contact p	Validation failure
VT-43	Click mode button that changes labels on products page	Error message
VT-46	Submit contact form with empty fields	error message
VT-48	Test error recovery on login with invalid email	error message clearance
VT-49	Clear contact form after filling it	error message
VT-51	Submit login form with empty fields	error message
VT-52	Test error recovery on contact form	error message clearance

3. Application Coverage Summary

Metric	Value
Pages Crawled	9
Pages with Interactive Elements	9
Input Fields Identified	9
Buttons Identified	18
Navigation Links	65
Dropdowns/Selects	0
Test Cases Generated	52
Validation Pass Rate	98.1%

4. Batch Generation Strategy

Tests were generated in 6 adversarial batches. Each batch focused on a distinct exploration category to guarantee structural diversity and maximize bug-finding potential.

Batch	Category	Focus
1	Input Torture and Field Poisoning	Edge-case values: long strings, special chars, min-1/max+1, empty required fields.
2	State Transitions and Form Abandonment	Partial fills, navigate-away-return, recalculate after change, state persistence.

3	Boundary Value Analysis	Exact min, max, zero, +1/-1 boundaries across every numeric input field.
4	Multi-Step Navigation Chaos	Fill on page A, jump to B, return to A -- does state survive?
5	Mode Switching and UI State Dances	Toggle modes mid-fill, switch tabs, check value retention after mode change.
6	Error Recovery and Validation Resilience	Submit empty, fix one field, resubmit; clear after success; recovery flow testing.

5. Test Coverage Classification

Category	Count
Authentication Tests	17
Boundary Tests	11
Negative Input Tests	9
General Exploratory Tests	8
Validation Enforcement Tests	3
Zero/Empty Input Tests	2
Navigation & Cross-Page Tests	2

Test cases are automatically classified based on their goals and step patterns. Categories are not mutually exclusive; each test is assigned to its primary category.

6. Validated Test Cases

52 test cases passed anti-hallucination validation. All referenced UI elements exist in the application snapshot.

Each test is marked PASSED (green) or FAILED (red) based on execution results.

VT-01 | Break login with empty password [FAILED - BUG FOUND]

Page: /login

Target: Enter password

Type: Authentication Tests

1. Navigate to '/login'
2. Enter " " into 'Enter password'
3. Click 'Sign In'

Expected: validation failure

!! Application did not meet expected behavior - potential bug detected.

VT-02 | Poison registration with long string [PASSED]

Page: /register

Target: Jane Doe

Type: Authentication Tests

1. Navigate to '/register'
2. Enter 'a' * 1000 into 'Jane Doe'
3. Click 'Create Account'

Expected: error message

VT-03 | Contaminate contact form with special characters [FAILED - BUG FOUND]

Page: /contact

Target: Your name

Type: Negative Input Tests

1. Navigate to '/contact'
2. Enter '!@#\$\$' into 'Your name'
3. Click 'Send Message'

*Expected: validation failure***!! Application did not meet expected behavior - potential bug detected.****VT-04 | Exceed max limit in registration [PASSED]**

Page: /register

Target: Min 8 characters

Type: Boundary Tests

1. Navigate to '/register'
2. Enter 'a' * 9 into 'Min 8 characters'
3. Click 'Create Account'

*Expected: error message***VT-05 | Break login with whitespace-only email [FAILED - BUG FOUND]**

Page: /login

Target: you@example.com

Type: Authentication Tests

1. Navigate to '/login'
2. Enter ' ' into 'you@example.com'
3. Click 'Sign In'

*Expected: validation failure***!! Application did not meet expected behavior - potential bug detected.****VT-06 | Cross-contaminate registration fields [PASSED]**

Page: /register

Target: Jane Doe

Type: Authentication Tests

1. Navigate to '/register'
2. Enter 'password123' into 'Jane Doe'
3. Enter 'password123' into 'you@example.com'
4. Click 'Create Account'

*Expected: error message***VT-07 | Poison contact form with dangerous string**

Page: /contact

Target: +1 (555) 000-0000

Type: Negative Input Tests

1. Navigate to '/contact'
2. Enter '<script>alert(1)</script>' into '+1 (555) 000-0000'
3. Click 'Send Message'

*Expected: validation failure***VT-08 | Break registration with empty required field [PASSED]**

Page: /register

Target: Jane Doe

Type: Validation Enforcement Tests

1. Navigate to '/register'
2. Enter '' into 'Jane Doe'
3. Click 'Create Account'

Expected: validation failure

VT-09 | Exceed min limit in contact form [FAILED - BUG FOUND]

Page: /contact

Target: +1 (555) 000-0000

Type: Boundary Tests

1. Navigate to '/contact'
2. Enter '-1' into '+1 (555) 000-0000'
3. Click 'Send Message'

*Expected: error message***!! Application did not meet expected behavior - potential bug detected.****VT-10 | Test form submission with empty email field [FAILED - BUG FOUND]**

Page: /login

Target: Enter password

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'password123' into 'Enter password'
3. Click 'Sign In'

*Expected: Validation failure for missing email***!! Application did not meet expected behavior - potential bug detected.****VT-11 | Test form submission with empty password field [FAILED - BUG FOUND]**

Page: /login

Target: you@example.com

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'user@example.com' into 'you@example.com'
3. Click 'Sign In'

*Expected: Validation failure for missing password***!! Application did not meet expected behavior - potential bug detected.****VT-12 | Test form submission with valid data and then resubmit [FAILED - BUG FOUND]**

Page: /login

Target: you@example.com

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'user@example.com' into 'you@example.com'
3. Enter 'password123' into 'Enter password'
4. Click 'Sign In'
5. Click 'Sign In'

*Expected: Validation failure or error message***!! Application did not meet expected behavior - potential bug detected.****VT-13 | Test form submission with partial data [PASSED]**

Page: /register

Target: Jane Doe

Type: Authentication Tests

1. Navigate to '/register'
2. Enter 'Jane Doe' into 'Jane Doe'
3. Click 'Create Account'

*Expected: Validation failure for missing fields***VT-14 | Test form submission with one empty field at a time [PASSED]**

Page: /register

Target: Jane Doe

Type: Boundary Tests

1. Navigate to '/register'
2. Enter 'Jane Doe' into 'Jane Doe'
3. Enter 'you@example.com' into 'you@example.com'
4. Enter 'password123' into 'Min 8 characters'

5. Click 'Create Account'

Expected: Validation failure for missing repeat password

VT-15 | Test form submission with valid data and then modify one fie [PASSED]

Page: /register

Target: Jane Doe

Type: Boundary Tests

1. Navigate to '/register'
2. Enter 'Jane Doe' into 'Jane Doe'
3. Enter 'you@example.com' into 'you@example.com'
4. Enter 'password123' into 'Min 8 characters'
5. Enter 'password123' into 'Repeat password'
6. Click 'Create Account'
7. Enter 'Jane Doe2' into 'Jane Doe'
8. Click 'Create Account'

Expected: Validation failure or error message

VT-16 | Test form submission with empty phone number field [FAILED - BUG FOUND]

Page: /contact

Target: Your name

Type: Zero/Empty Input Tests

1. Navigate to '/contact'
2. Enter 'Your Name' into 'Your name'
3. Enter 'you@example.com' into 'you@example.com'
4. Click 'Send Message'

Expected: Validation failure for missing phone number

!! Application did not meet expected behavior - potential bug detected.

VT-17 | Test form submission with empty name field [FAILED - BUG FOUND]

Page: /contact

Target: +1 (555) 000-0000

Type: Negative Input Tests

1. Navigate to '/contact'
2. Enter '+1 (555) 000-0000' into '+1 (555) 000-0000'
3. Enter 'you@example.com' into 'you@example.com'
4. Click 'Send Message'

Expected: Validation failure for missing name

!! Application did not meet expected behavior - potential bug detected.

VT-18 | Test form submission with empty email field [FAILED - BUG FOUND]

Page: /contact

Target: Your name

Type: Negative Input Tests

1. Navigate to '/contact'
2. Enter 'Your Name' into 'Your name'
3. Enter '+1 (555) 000-0000' into '+1 (555) 000-0000'
4. Click 'Send Message'

Expected: Validation failure for missing email

!! Application did not meet expected behavior - potential bug detected.

VT-19 | Break the login page with minimum password length [FAILED - BUG FOUND]

Page: /login

Target: Min 8 characters

Type: Boundary Tests

1. Navigate to '/login'
2. Enter 'a' into 'Min 8 characters'
3. Click 'Sign In'

Expected: Password is too short error message

!! Application did not meet expected behavior - potential bug detected.

VT-20 | Crash the register page with very large name input [PASSED]

Page: /register

Target: Jane Doe

Type: Boundary Tests

1. Navigate to '/register'
2. Enter '9999999999' into 'Jane Doe'
3. Click 'Create Account'

Expected: Invalid input error message

VT-21 | Test the contact page with zero phone number [FAILED - BUG FOUND]

Page: /contact

Target: +1 (555) 000-0000

Type: Boundary Tests

1. Navigate to '/contact'
2. Enter '0' into '+1 (555) 000-0000'
3. Click 'Send Message'

Expected: Invalid phone number error message

!! Application did not meet expected behavior - potential bug detected.

VT-22 | Break the products page with non-numeric input [PASSED]

Page: /products

Target: N/A

Type: Negative Input Tests

1. Navigate to '/products'
2. This step is not possible as there are no input fields on this page, so we will skip it

Expected: No input fields available

VT-23 | Test the login page with maximum allowed password length [FAILED - BUG FOUND]

Page: /login

Target: Enter password

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'a'.repeat(1000) into 'Enter password'
3. Click 'Sign In'

Expected: Password is too long error message

!! Application did not meet expected behavior - potential bug detected.

VT-24 | Crash the register page with very small email input [PASSED]

Page: /register

Target: you@example.com

Type: Authentication Tests

1. Navigate to '/register'
2. Enter 'a@b.c' into 'you@example.com'
3. Click 'Create Account'

Expected: Invalid email error message

VT-25 | Break the login page with one above maximum password length [FAILED - BUG FOUND]

Page: /login

Target: Enter password

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'a'.repeat(1001) into 'Enter password'
3. Click 'Sign In'

Expected: Password is too long error message

!! Application did not meet expected behavior - potential bug detected.

VT-26 | Test the register page with zero name input [PASSED]

Page: /register

Target: Jane Doe

Type: Boundary Tests

1. Navigate to '/register'
2. Enter '0' into 'Jane Doe'
3. Click 'Create Account'

Expected: Invalid name error message

VT-27 | Test rapid form submission on login page [FAILED - BUG FOUND]

Page: /login

Target: you@example.com

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'test@example.com' into 'you@example.com'
3. Enter 'password123' into 'Enter password'
4. Click 'Sign In'
5. Click 'Sign In' again

Expected: Validation failure due to duplicate submission

!! Application did not meet expected behavior - potential bug detected.

VT-28 | Test form submission with empty optional fields on contact p [FAILED - BUG FOUND]

Page: /contact

Target: Your name

Type: Zero/Empty Input Tests

1. Navigate to '/contact'
2. Enter 'John Doe' into 'Your name'
3. Enter 'john@example.com' into 'you@example.com'
4. Click 'Send Message'

Expected: Error message for missing phone number

!! Application did not meet expected behavior - potential bug detected.

VT-29 | Test rapid input filling on register page [PASSED]

Page: /register

Target: Jane Doe

Type: Boundary Tests

1. Navigate to '/register'
2. Enter 'Jane Doe' into 'Jane Doe'
3. Enter 'jane@example.com' into 'you@example.com'
4. Enter 'password123' into 'Min 8 characters'
5. Enter 'password123' into 'Repeat password'
6. Click 'Create Account'

Expected: No UI errors or crashes

VT-30 | Test form submission with cleared fields on login page [FAILED - BUG FOUND]

Page: /login

Target: you@example.com

Type: Authentication Tests

1. Navigate to '/login'
2. Enter 'test@example.com' into 'you@example.com'
3. Enter 'password123' into 'Enter password'
4. Click 'Dismiss'
5. Enter 'new@example.com' into 'you@example.com'
6. Enter 'newpassword' into 'Enter password'
7. Click 'Sign In'

Expected: Validation failure due to incorrect credentials

!! Application did not meet expected behavior - potential bug detected.

VT-31 | Test multiple submit clicks on products page [PASSED]

Page: /products

Target: N/A

Type: General Exploratory Tests

- 1. Navigate to '/products'
- 2. Click 'Add Widget Pro'
- 3. Click 'Add Widget Pro' again
- 4. Click 'Order'

Expected: Error message for duplicate order submission

VT-32 | Test rapid form submission on register page [PASSED]

Page: /register

Target: Jane Doe

Type: Boundary Tests

- 1. Navigate to '/register'
- 2. Enter 'Jane Doe' into 'Jane Doe'
- 3. Enter 'jane@example.com' into 'you@example.com'
- 4. Enter 'password123' into 'Min 8 characters'
- 5. Enter 'password123' into 'Repeat password'
- 6. Click 'Create Account'
- 7. Click 'Create Account' again

Expected: Validation failure due to duplicate submission

VT-33 | Test multiple submit clicks on contact page [FAILED - BUG FOUND]

Page: /contact

Target: Your name

Type: Negative Input Tests

- 1. Navigate to '/contact'
- 2. Enter 'John Doe' into 'Your name'
- 3. Enter 'john@example.com' into 'you@example.com'
- 4. Enter '+1 (555) 000-0000' into '+1 (555) 000-0000'
- 5. Click 'Send Message'
- 6. Click 'Send Message' again

Expected: Error message for duplicate submission

!! Application did not meet expected behavior - potential bug detected.

VT-34 | Test form submission with rapid input filling on login page [FAILED - BUG FOUND]

Page: /login

Target: you@example.com

Type: Authentication Tests

- 1. Navigate to '/login'
- 2. Enter 'test@example.com' into 'you@example.com'
- 3. Enter 'password123' into 'Enter password'
- 4. Click 'Sign In'

Expected: Validation failure due to incorrect credentials

!! Application did not meet expected behavior - potential bug detected.

VT-35 | Test mode switching on products page [PASSED]

Page: /products

Target: N/A

Type: General Exploratory Tests

- 1. Navigate to '/products'
- 2. Click 'Add Widget Pro'
- 3. Click 'Add Gadget Lite'
- 4. Click 'Add Component X'

Expected: No UI errors

VT-36 | Rapidly apply and unapply mode on dashboard page [PASSED]

Page: /dashboard

Target: N/A

Type: Authentication Tests

- 1. Navigate to '/dashboard'
- 2. Click 'Account v'
- 3. Click 'Account v'
- 4. Click 'Dismiss'

Expected: Validation failure

VT-37 | Switch between modes on products page [PASSED]

Page: /products

Target: N/A

Type: Navigation & Cross-Page Tests

- 1. Navigate to '/products'
- 2. Click 'Show Special Offers'
- 3. Click 'Order'
- 4. Click 'Show Special Offers'

Expected: Error message

VT-38 | Test mode switching with incomplete fields on contact page [FAILED - BUG FOUND]

Page: /contact

Target: Your name

Type: General Exploratory Tests

- 1. Navigate to '/contact'
- 2. Click 'Send Message'
- 3. Enter 'value' into 'Your name'
- 4. Click 'Send Message'

Expected: Validation failure

!! Application did not meet expected behavior - potential bug detected.

VT-39 | Click mode button multiple times on products page [PASSED]

Page: /products

Target: N/A

Type: General Exploratory Tests

- 1. Navigate to '/products'
- 2. Click 'Add Widget Pro'
- 3. Click 'Add Widget Pro'
- 4. Click 'Add Widget Pro'

Expected: No UI errors

VT-40 | Test mode switching with different labels on products page [PASSED]

Page: /products

Target: N/A

Type: General Exploratory Tests

- 1. Navigate to '/products'
- 2. Click 'Add Gadget Lite'
- 3. Click 'Add Component X'
- 4. Click 'Order'

Expected: Error message

VT-41 | Rapidly switch between modes on dashboard page [PASSED]

Page: /dashboard

Target: N/A

Type: Navigation & Cross-Page Tests

- 1. Navigate to '/dashboard'
- 2. Click 'Account v'
- 3. Click 'Dismiss'
- 4. Click 'Account v'

Expected: Validation failure

VT-42 | Test mode switching with incomplete fields on register page [PASSED]

Page: /registerTarget: Jane DoeType: Authentication Tests

1. Navigate to '/register'

2. Click 'Create Account'

3. Enter 'value' into 'Jane Doe'

4. Click 'Create Account'

Expected: Validation failure

VT-43 | Click mode button that changes labels on products page [FAILED - BUG FOUND]

Page: /productsTarget: N/AType: General Exploratory Tests

1. Navigate to '/products'

2. Click 'Show Special Offers'

3. Click 'Order'

4. Click 'Add Widget Pro'

Expected: Error message

!! Application did not meet expected behavior - potential bug detected.

VT-44 | Test mode switching with rapid mode changes on products page [PASSED]

Page: /productsTarget: N/AType: General Exploratory Tests

1. Navigate to '/products'

2. Click 'Add Widget Pro'

3. Click 'Add Gadget Lite'

4. Click 'Add Component X'

5. Click 'Add Widget Pro'

Expected: No UI errors

VT-45 | Test register with invalid repeat password [PASSED]

Page: /registerTarget: Min 8 charactersType: Boundary Tests

1. Navigate to '/register'

2. Enter 'password123' into 'Min 8 characters'

3. Enter 'wrongpassword' into 'Repeat password'

4. Click 'Create Account'

Expected: error message

VT-46 | Submit contact form with empty fields [FAILED - BUG FOUND]

Page: /contactTarget: Your nameType: Validation Enforcement Tests

1. Navigate to '/contact'

2. Enter " into 'Your name'

3. Enter " into 'you@example.com'

4. Enter " into '+1 (555) 000-0000'

5. Click 'Send Message'

Expected: error message

!! Application did not meet expected behavior - potential bug detected.

VT-47 | Order products without selecting any [PASSED]

Page: /products

Target: N/A

Type: General Exploratory Tests

- 1. Navigate to '/products'
- 2. Click 'Order'

Expected: error message

VT-48 | Test error recovery on login with invalid email [FAILED - BUG FOUND]

Page: /login

Target: you@example.com

Type: Negative Input Tests

- 1. Navigate to '/login'
- 2. Enter 'invalidemail' into 'you@example.com'
- 3. Enter 'password123' into 'Enter password'
- 4. Click 'Sign In'
- 5. Enter 'you@example.com' into 'you@example.com'
- 6. Click 'Sign In'

Expected: error message clearance

!! Application did not meet expected behavior - potential bug detected.

VT-49 | Clear contact form after filling it [FAILED - BUG FOUND]

Page: /contact

Target: Your name

Type: Negative Input Tests

- 1. Navigate to '/contact'
- 2. Enter 'John Doe' into 'Your name'
- 3. Enter 'you@example.com' into 'you@example.com'
- 4. Enter '+1 (555) 000-0000' into '+1 (555) 000-0000'
- 5. Click 'Dismiss'
- 6. Click 'Send Message'

Expected: error message

!! Application did not meet expected behavior - potential bug detected.

VT-50 | Test register with empty fields [PASSED]

Page: /register

Target: N/A

Type: Authentication Tests

- 1. Navigate to '/register'
- 2. Click 'Create Account'

Expected: error message

VT-51 | Submit login form with empty fields [FAILED - BUG FOUND]

Page: /login

Target: N/A

Type: Validation Enforcement Tests

- 1. Navigate to '/login'
- 2. Click 'Sign In'

Expected: error message

!! Application did not meet expected behavior - potential bug detected.

VT-52 | Test error recovery on contact form [FAILED - BUG FOUND]

Page: /contact

Target: you@example.com

Type: Negative Input Tests

- 1. Navigate to '/contact'
- 2. Enter 'invalidemail' into 'you@example.com'
- 3. Enter '+1 (555) 000-0000' into '+1 (555) 000-0000'
- 4. Click 'Send Message'

5. Enter 'you@example.com' into 'you@example.com'

6. Click 'Send Message'
- Expected: error message clearance
- !! Application did not meet expected behavior - potential bug detected.

7. Structural Validation Confirmation

All 52 test cases were validated against the UI exploration snapshot. The validation engine verified that:

- ✓ All referenced UI element labels exist in the application snapshot.

✓ All referenced page paths exist in the crawled page inventory.

✓ No hallucinated or invented element labels were detected.

✓ Each test case has a valid structure: ID, goal, steps (min 2), and expected result.
- 1 test case(s) were rejected during validation.
- ET-60: Too few steps (minimum 2)

8. Appendix

Snapshot Statistics

Metric	Value
Snapshot File	ai_exploration_snapshot.json
Snapshot Size	20912 bytes
Total Pages in Snapshot	9
Total Input Fields	9
Total Buttons	18
Total Navigation Links	65

Pipeline Configuration

Parameter	Value
AI Model	llama-3.3-70b-versatile (Groq)
Generation Batches	6
Tests per Batch	~10
Batch Delay	30 seconds
Validation Engine	3-Layer Anti-Hallucination
Step Generator	Deterministic (Template-Based)

Output Files Reference

File	Path
Raw Test Cases	data/generated_test_cases.json
Validated Test Cases	data/validated_test_cases.json
Rejected Test Cases	data/rejected_test_cases.json
Gauge Specification	specs/ai_exploration.spec
Step Implementations	step_impl/step_implementation.py
This Report	reports/test_report.pdf